

# LOU BAKER, PhD

## SUMMARY

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Educator and researcher with expertise in image processing, data analysis, and experimental fluid mechanics. Dedicated to teaching, mentorship, and improving equity and accessibility in academia.

## EDUCATION

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### University of Minnesota, Minneapolis, MN

**Ph.D.**, Aerospace Engineering and Mechanics | 4.00 Jun 2021

*Focus: Experimental Fluid Mechanics*

**M.S.**, Aerospace Engineering and Mechanics | 4.00 Dec 2018

**B.Eng.**, Aerospace Engineering and Mechanics | 3.75 May 2016

## EXPERIENCE

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**Financial Secretary & Field Organizer**, UAW Local 4121, Seattle, WA Aug 2023 – present  
Organizing for equity and accessibility for academic workers at the University of Washington.  
Managing the Local's staff, finances, and database.

### Teaching

**Part-time Lecturer**, Northeastern University, Seattle, WA May – Jun 2025  
ME 3465: Introduction to Flight

**Guest Lecturer**, University of Washington, Seattle, WA May 2022  
ME 543: Fluid Turbulence

**Laboratory Instructor**, University of Minnesota, Minneapolis, MN Sep 2016 – May 2017  
AEM 4601: Instrumentation Laboratory  
AEM 4602: Aeromechanics Laboratory

**Teaching Assistant**, University of Minnesota, Minneapolis, MN Jan – May 2014  
AEM 2012: Dynamics

### Research

**NSF Postdoctoral Fellow**, University of Washington, Seattle, WA Jun 2021 – Jul 2023  
*Environmental Fluid Mechanics Group, Mechanical Engineering Dept*  
Investigated microplastic particle dynamics in wavy, wind-driven surface flows

**Graduate Research Assistant**, University of Minnesota, Minneapolis, MN May 2016 – Jun 2021  
*St. Anthony Falls Laboratory, Aerospace Engineering & Mechanics Dept*  
Explored fluid-particle dynamics in turbulent open channel flows

**Undergraduate Research Assistant**, University of Minnesota Jan 2014 – May 2016  
*Aerospace Engineering & Mechanics Dept*  
Studied particle clustering in homogeneous, isotropic turbulence

## LOU BAKER, PhD

- Intern**, Virgin Orbit, CA Jun – Aug 2015  
Designed engine gimbals and heat exchangers for LauncherOne rocket engines
- Intern**, NASA Jet Propulsion Laboratory, CA Jun – Aug 2013 & 2014  
Designed and tested magnetic shielding for the ISS Cold Atom Lab research facility

## GRANTS & AWARDS

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- NSF Ocean Sciences Postdoctoral Research Fellowship** Aug 2021 – Aug 2023
- National Defense Science and Engineering Graduate Fellowship** Sep 2017 – Jun 2021
- Edward Silberman Fellowship** Jan 2019 – May 2020
- John A. & Jane Dunning Copper Fellowship** May 2017
- Albert George Oswald Prize for Outstanding Research** Sep 2015 – May 2016

## PUBLICATIONS

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9. **Baker LJ**, Chávez-Dorado JE, Aggarwal A, Garrey I, & DiBenedetto MH. “The importance of particle shape on sunlight exposure in a wind-driven, free-surface flow.” *In preparation*.
8. Sanness Salmon H, **Baker LJ**, Kozarek J, & Coletti F (2023) “Effect of Shape and Size on the Transport of Floating Particles on the Free-surface of a Meandering Stream.” *Water Resources Research*, 59, e2023WR035716.
7. **Baker LJ** & DiBenedetto MH (2023) “Large-scale particle shadow tracking and orientation measurement with collimated light.” *Experiments in Fluids*, 64, 52.
6. **Baker LJ** & Coletti F (2022) “Experimental investigation of inertial fibres and disks in a turbulent boundary layer.” *Journal of Fluid Mechanics*, 943, A27.
5. **Baker LJ**, Qiao Y, Ghaemi S, & Coletti F (2021) “Method to minimize polymer degradation in drag-reduced non-Newtonian turbulent boundary layers.” *Measurement Science and Technology*, 32, 085303.
4. **Baker LJ** & Coletti F (2021) “Particle–fluid–wall interaction of inertial spherical particles in a turbulent boundary layer.” *Journal of Fluid Mechanics*, 908, A39.
3. **Baker LJ** & Coletti F (2019) “Experimental study of negatively-buoyant finite-size particles in a turbulent boundary layer up to dense regimes.” *Journal of Fluid Mechanics*, 866, 598-629.
2. Petersen A, **Baker LJ**, & Coletti F (2019) “Experimental study of inertial particles clustering and settling in homogeneous turbulence.” *Journal of Fluid Mechanics*, 864, 925-970.
1. **Baker LJ**, Frankel A, Mani A, & Coletti F (2017) “Coherent clusters of inertial particles in homogeneous turbulence.” *Journal of Fluid Mechanics*, 833, 364-398.

## INVITED TALKS

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- Mechanical Engineering Graduate Seminar**, University of Iowa (Feb 2023)
- Workshop on Ocean Microplastic Transport**, Banff International Research Station (Feb 2022)

## LOU BAKER, PhD

**Ocean Engineering Seminar**, University of New Hampshire (Feb 2022)

**Hydro-Geo Seminar**, Univ. of Minnesota & Univ. of Illinois Urbana-Champaign (Feb 2022)

**Environmental Fluid Mechanics Seminar**, University of Washington (Oct 2021)

## MENTORSHIP, OUTREACH, & SERVICE

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### Peer Review

Mar 2019 – present

Journal referee: *Journal of Fluid Mechanics*, *Experiments in Fluids*, *International Communications on Heat and Mass Transfer*, *European Journal of Mechanics / B Fluids*, *Physical Review Fluids*, *Water Research*, *Microplastics & Nanoplastics*, *Journal of Geophysical Research - Solid Earth*

Proposal reviewer: NSF

**Research Mentor**, University of Washington

Jun 2021 – Jan 2023

**Postdoc Union Steward**, University of Washington

May 2022 – Jul 2023

**Climate Justice Working Group**, University of Washington

Jun 2021 – Jul 2022

**Volunteer Income Tax Preparer**, United Way of King County, WA

Jan 2023 – Apr 2025

**Outreach Volunteer**, Seattle Astronomical Society

Feb 2022 – Jan 2023

**Peer Mentor**, Women of Aeronautics and Astronautics, Univ. of Minnesota

Sep 2017 – Jun 2021

## PROFESSIONAL DEVELOPMENT

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**Leadership Academy and Network for Diversity in the Geosciences**, American Geophysical

Union Postdoctoral Research Fellows Program

Jan 2022 – Jan 2023

**Space Weather Simulation Summer School**, University of Colorado

Jul 2022

**Summer Institute on Earth-Surface Dynamics**, St. Anthony Falls Lab

Aug 2017

**Solar Vehicle Project**, University of Minnesota

Sep 2012 – Jan 2015

## CONFERENCE PRESENTATIONS

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Chávez-Dorado JE, Riley J, **Baker LJ**, & DiBenedetto MH (2026) "Lagrangian dispersion of finite-sized particles under surface waves and turbulence." APS Division of Fluid Dynamics, Orlando, FL.

Chávez-Dorado JE, **Baker LJ**, & DiBenedetto MH (2026) "Buoyant Particle Transport in Free-Surface Boundary Layers." AGU Ocean Sciences Meeting, Glasgow, Scotland.

Chávez-Dorado JE, **Baker LJ**, & DiBenedetto MH (2025) "Vertical transport of buoyant particles under a free-surface boundary layer." APS Division of Fluid Dynamics, Houston, TX.

Chávez-Dorado JE, Maschhoff A, **Baker LJ**, & DiBenedetto MH (2024) "Particle transport under a wind-driven boundary layer." APS Division of Fluid Dynamics, Salt Lake City, UT.

DiBenedetto MH, Chávez-Dorado JE, & **Baker LJ** (2024) "Dynamics of buoyant, non-spherical particles in a wind-driven surface boundary layer." Ocean Sciences Meeting, New Orleans, LA.

**Baker LJ**, Aggarwal A, Chávez-Dorado JE, Garrey I, & DiBenedetto MH (2023) "Non-spherical particle orientation dynamics under a wind-driven free surface boundary layer." APS Division of Fluid Dynamics, Washington D.C.

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- Baker LJ**, Aggarwal A, Chávez-Dorado JE, Garrey I, & DiBenedetto MH (2023) "Buoyant, Non-Spherical Particles in Wind-Driven Waves." Waves in Sea Environment Meeting, Princeton, NJ.
- Baker, L**, Aggarwal A, Chávez-Dorado JE, Garrey I, & DiBenedetto MH (2022) "Buoyant, Non-Spherical Plastic Particles in Turbulent Wind-Driven Waves." AGU Fall Meeting, Chicago, IL.
- Baker LJ**, Aggarwal A, Chávez-Dorado JE, Garrey I, & DiBenedetto MH (2022) "Buoyant, non-spherical particles in turbulent wind-driven waves." APS Division of Fluid Dynamics, Indianapolis, IN.
- Baker LJ**, Aggarwal A, Chávez-Dorado JE, Garrey I, & DiBenedetto MH (2022) "Buoyant, non-spherical particles in turbulent wind-driven waves." Atmospheres, Oceans, Earths—Unifying perspectives on geophysical and environmental multiphase flows, KITP, Santa Barbara, CA.
- Axworthy J, Wang S, DiBenedetto MH, **Baker LJ**, & Padilla-Gamiño J (2022) "Microplastics ingestion and adhesion by reef-building corals under different flow rates." International Coral Reef Society, Bremen, Germany.
- Baker LJ** & DiBenedetto MH (2022) "Vertical transport and orientation of buoyant, non-spherical particles in the wind-mixed ocean surface boundary layer." Ocean Sciences Meeting, virtual.
- Baker LJ** & Coletti F (2021) "Orientation and tumbling of inertial rod and disk particles in a turbulent boundary layer." APS Division of Fluid Dynamics, Phoenix, AZ.
- Baker LJ** & Coletti F (2020) "Effects of shape on microplastic particle–fluid–wall interaction and transport in a turbulent boundary layer." AGU Fall Meeting, virtual.
- Baker LJ** & Coletti F (2020) "Particle-fluid-wall interaction of anisotropic inertial particles in a turbulent boundary layer." APS Division of Fluid Dynamics, virtual.
- Sanness Salmon H, **Baker LJ**, Kozarek J, & Coletti F (2020) "Effect of size and shape on the transport of particles over the free surface of a natural stream." APS Division of Fluid Dynamics, virtual.
- Baker LJ** & Coletti F (2019) "Experimental Investigation of the Dynamics of Resuspending Spherical Sediment Particles in a Turbulent Boundary Layer." AGU Fall Meeting, San Francisco, CA.
- Coletti F & **Baker LJ** (2019) "Simultaneous tracking of suspended particles and time-resolved PIV in a turbulent boundary layer." APS Division of Fluid Dynamics, Seattle, WA.
- Petersen A, **Baker LJ**, & Coletti F (2017) "Laboratory Study of Air Turbulence-Particle Coupling." AGU Fall Meeting, New Orleans, LA.
- Baker LJ** & Coletti F (2017) "Experimental study of dense suspension of large particles in a turbulent boundary layer." APS Division of Fluid Dynamics, Denver, CO.
- Petersen A, **Baker LJ**, & Coletti F (2017) "Particle Plumes Falling Through Quiescent and Turbulent Environments." APS Division of Fluid Dynamics, Denver, CO.
- Petersen A, Carter D, **Baker LJ**, & Coletti F (2017) "Experimental Study of Particle-turbulence Interaction in Homogeneous Turbulence." 10th International Symposium on Turbulence and Shear Flow Phenomena, Chicago, IL, USA.
- Coletti F, Toloui M, Fong, KO, Nemes A, & **Baker LJ** (2016) "Volumetric distribution and velocity of inertial particles in a turbulent channel flow." 18th International Symposium on Application of Laser and Imaging Techniques to Fluid Mechanics, Lisbon, Portugal.
- Coletti F, Petersen A, Carter D, & **Baker LJ** (2016) "Measurements of particle settling velocity in homogeneous turbulence with no mean flow." International Conference on Multiphase Flows, Florence, Italy.
- Baker LJ**, Frankel A, Mani A, & Coletti F (2016) "Coherent clusters of inertial particles in homogeneous turbulence." APS Division of Fluid Dynamics, Portland, OR.
- Petersen A, Carter D, **Baker LJ**, & Coletti F (2015) "Settling of inertial particles through quiescent, weakly turbulent and strongly turbulent air." APS Division of Fluid Dynamics, Boston, MA.